Hamming Weight Scaling of Grover's Algorithm under Amplitude Damping Noise

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Abstract

We investigate the behavior of Grover's quantum search algorithm under amplitude damping noise in the near-term (NISQ) regime. Through systematic simulations on 2–4 qubit systems, we observe that the algorithmic success probability follows an exponential decay law,

$$P_{\text{success}} \propto \exp(-pwd),$$

where p is the amplitude damping strength, w is the Hamming weight, and d is the circuit depth.

Motivated by this dependence, we explore whether encoding strategies that reduce the Hamming weight of the marked state can mitigate noise-induced degradation. Counter to naive expectations, such encodings provide no measurable improvement. Using time-resolved density matrix tomography, we show that Grover's algorithm maintains an average Hamming weight $\langle w(t) \rangle \approx n/2$ throughout most of its execution, largely independent of the target state or encoding.

These results indicate that time-averaged Hamming weight, rather than the final marked state alone, governs sensitivity to amplitude damping. We conjecture that this mechanism may represent a general limitation for amplitude amplification algorithms operating under directional noise, with implications for the practical viability of quantum search on NISQ devices.

1 Introduction

Grover's quantum search algorithm provides a quadratic speedup over classical unstructured search, locating a marked element among $N = 2^n$ possibilities using $O(\sqrt{N})$ oracle queries. This result underpins a wide range of quantum algorithms, including amplitude amplification, quantum optimization, and search-based primitives in cryptography and machine learning.

Despite its asymptotic advantage, the practical realization of Grover's algorithm on near-term quantum hardware remains challenging. Current devices operate in the noisy intermediate-scale quantum (NISQ) regime, where decoherence and gate errors accumulate over circuit depth. While prior work has studied the general impact of noise on quantum algorithms, the specific mechanisms by which different noise channels degrade Grover's performance are not yet fully understood.

In this work, we focus on amplitude damping noise, which models energy relaxation (T_1 processes) and exhibits a strong directional bias in the computational basis. We demonstrate that under amplitude damping, Grover's success probability depends exponentially on both circuit depth and the Hamming weight of the quantum state. We further show that attempts to reduce this sensitivity via state encoding fail due to the intrinsic structure of the algorithm's quantum trajectory.

2 Background

2.1 Grover's Algorithm

Grover's algorithm operates on n qubits initialized in the uniform superposition

$$|s\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle. \tag{1}$$

An oracle O applies a phase flip to a marked basis state $|\omega\rangle$,

$$O|\omega\rangle = -|\omega\rangle, \quad O|x\rangle = |x\rangle \;\; (x \neq \omega), \tag{2}$$

followed by the diffusion operator

$$D = 2|s\rangle\langle s| - I. \tag{3}$$

Repeated application of DO rotates the state toward $|\omega\rangle$, achieving optimal success probability after $O(\sqrt{2^n})$ iterations in the noiseless setting.

2.2 Amplitude Damping Noise

Amplitude damping models irreversible energy relaxation from $|1\rangle$ to $|0\rangle$ with probability p. It is described by Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \tag{4}$$

Unlike depolarizing or dephasing noise, amplitude damping is asymmetric and basis-dependent, preferentially suppressing excited-state population.

2.3 Hamming Weight

The Hamming weight $w(|x\rangle)$ of a computational basis state is defined as the number of qubits in the $|1\rangle$ state. For a mixed quantum state ρ, the expected Hamming weight is

$$\langle w \rangle = \sum_x \mathrm{Tr}(\rho |x\rangle\langle x|)\, w(|x\rangle). \tag{5}$$

Since amplitude damping acts independently on each excited qubit, the Hamming weight plays a central role in determining noise sensitivity.

3 Experimental Setup

All experiments were performed using Qiskit Aer simulations. Grover's algorithm was implemented for 2-, 3-, and 4-qubit systems with appropriate numbers of Grover iterations. Amplitude damping noise with strength $p = 0.1$ was applied after each single-qubit gate using Kraus operators.

To isolate the role of circuit depth independently of logical structure, we employed depth inflation by replacing each Hadamard gate H with H^3, which is logically equivalent since $H^2 = I$. Additional noise channels (phase damping, two-qubit depolarizing noise, and readout error) were included for comparison.

4 Results

4.1 Hamming Weight Dependence

Table 1 summarizes success probabilities for Grover's algorithm targeting states of increasing Hamming weight under amplitude damping.

Table 1: Success probabilities under 10% amplitude damping

Target State	Hamming Weight	P_{success}
$\|11\rangle$	2	0.71
$\|111\rangle$	3	0.23
$\|1111\rangle$	4	0.084

The decay rate increases systematically with Hamming weight, consistent with independent relaxation channels acting on each excited qubit.

4.2 Depth Scaling Law

Fitting the data to an exponential model yields

$$P_{\text{success}}(w, d) \approx \exp(-pwd), \tag{6}$$

which is consistent with a survival probability $(1-p)^{wd}$ for w excited qubits evolving over depth d.

4.3 Comparison of Noise Channels

A direct comparison between targets $|000\rangle$ and $|111\rangle$ under identical depth inflation reveals a strong asymmetry for amplitude damping but negligible differences for symmetric noise channels, confirming that the observed effect is specific to directional relaxation.

4.4 Encoding Experiments

We tested whether reducing the Hamming weight of the marked state via reversible encoding could improve performance. Despite altering the marked state's Hamming weight from $w = 4$ to $w = 1$, no statistically significant improvement was observed in the final success probability.

4.5 Time-Resolved Hamming Weight Dynamics

Density matrix tomography performed at intermediate circuit depths reveals that all encoding strategies maintain an average Hamming weight $\langle w(t) \rangle \approx n/2$ throughout most of the algorithm. The trajectories are nearly identical, indicating that the algorithm's superposition structure dominates noise exposure.

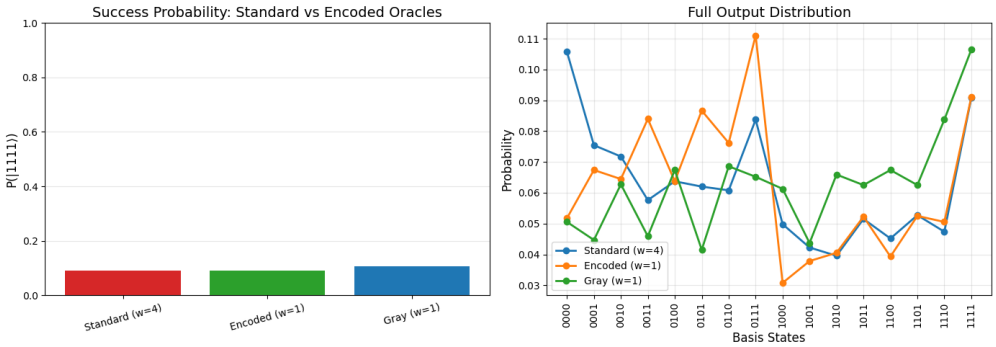

Figure 1: Comparison of standard and encoded oracle implementations. (a) Success probability of the marked state for standard and encoded Grover circuits. (b) Full output probability distribution over all computational basis states.

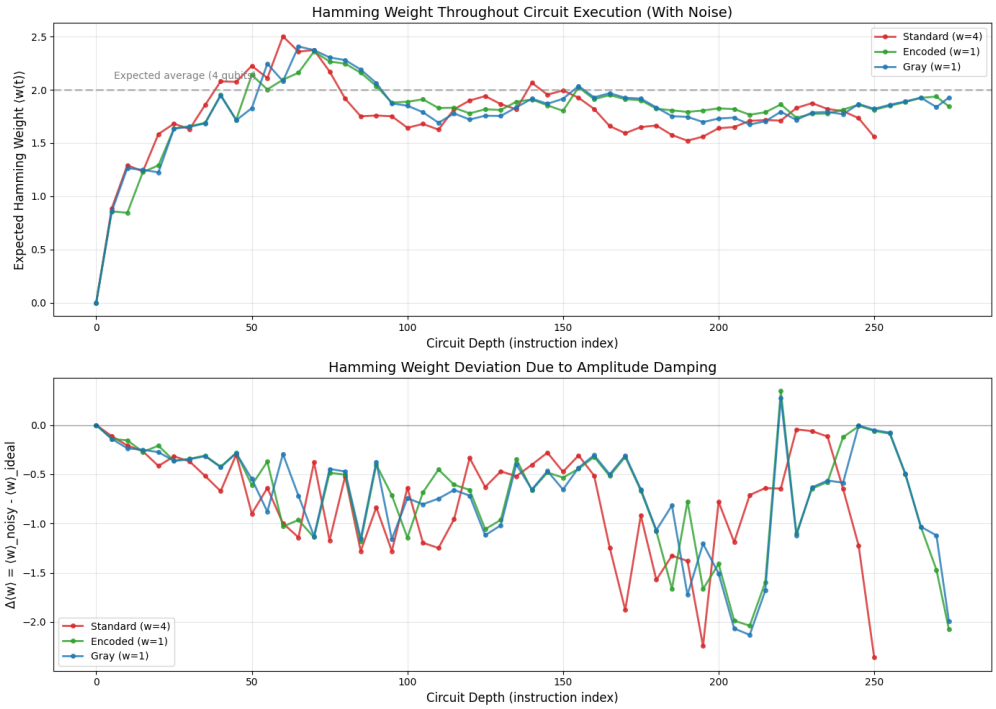

Figure 2: Time-resolved Hamming weight dynamics under amplitude damping. (a) Expected Hamming weight as a function of circuit depth during Grover iterations. (b) Overall suppression of excited-state population due to amplitude damping during amplification.

5 Analysis

Grover's algorithm evolves within a two-dimensional subspace spanned by the marked state and the uniform superposition of unmarked states. While the marked state may have low Hamming weight, the orthogonal component has expected Hamming weight $n/2$. Since this component dominates for most of the algorithm's runtime, the time-averaged Hamming weight remains largely independent of encoding.

This leads naturally to a refined decay model,

$$P_{\text{success}} \propto \exp\left(-p \int_0^d \langle w(t) \rangle \, dt\right) \approx \exp(-p\langle w_{\text{avg}} \rangle d), \tag{7}$$

with $\langle w_{\text{avg}} \rangle \approx n/2$ for Grover's algorithm.

6 Conjecture

Conjecture 1 (Hamming Weight Barrier)**.** *Any amplitude amplification algorithm that begins from a uniform superposition and achieves constant success probability must maintain a time-averaged Hamming weight $\langle w_{\text{avg}} \rangle \geq \alpha n$ for some constant $\alpha > 0$, over a circuit depth sufficient for amplification.*

This conjecture is motivated by empirical evidence and heuristic arguments rather than a formal proof.

7 Discussion

Our results suggest that amplitude damping constitutes a particularly severe limitation for quantum search on NISQ devices. Unlike symmetric noise channels, directional relaxation couples directly to the algorithm's population dynamics rather than its logical correctness.

These findings emphasize the importance of considering noise structure—not merely noise strength—when assessing the feasibility of quantum advantage.

8 Conclusion

We have shown that Grover's algorithm under amplitude damping exhibits exponential sensitivity to the product of circuit depth and time-averaged Hamming weight. Encoding strategies that reduce the Hamming weight of the marked state do not mitigate this effect due

to the algorithm's intrinsic superposition dynamics. This behavior raises fundamental questions about the robustness of amplitude amplification in the presence of directional noise and motivates further theoretical and experimental investigation.

References

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